Multivariate analysis project 3
Clustering

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Dataset Source:
This data is taken from the English Football Association. It was collected and analyzed by Opta.

Dataset description:

The data has 687 observations and 12 variables. Here's a description for each variable based on the Premier League 2021/2022 season:

• Player – The full name of the footballer (e.g., Mohamed Salah, Kevin De Bruyne).

• Team – The club the player represented during the season (e.g., Manchester City, Liverpool).

• Nation – The nationality of the player, usually represented by their FIFA-recognized country (e.g., England, Brazil, Portugal).

• Pos (Position) – The player's primary role on the pitch, typically categorized as GK (Goalkeeper), DF (Defender), MF (Midfielder), or FW (Forward).

• Age – The player's age in years as of a specific date (often the start of the season).

• MP (Matches Played) – The number of games the player appeared in, including both starts and substitute appearances.

• Starts – The number of matches in which the player was in the starting XI.

• Min (Minutes Played) – The total number of minutes the player was on the field during the season.

• Gls (Goals Scored) – The total number of goals the player scored in the Premier League that season.

• Ast (Assists) – The number of assists provided by the player, contributing to a teammate's goal.

• CrdY (Yellow Cards) – The total number of yellow cards received by the player.

• CrdR (Red Cards) – The total number of red cards received by the player, either directly or as a result of two yellow cards.

Project goals:

The goal of this project is to apply various clustering techniques to a football dataset (EPL Dataset.csv), and evaluate the quality of these clusterings using the R^2 statistic. R^2, in this context, represents the proportion of the total variance (TSS) explained by the clustering (1 - WSS/TSS), where WSS is the within-cluster sum of squares. This metric acts as a goodness-of-fit indicator and helps assess how well the clustering captures the structure in the data.

Data preparation:

The dataset was loaded and only numeric columns relevant to player performance were selected: Age, MP (Matches Played), Starts, Min (Minutes), Gls (Goals), Ast (Assists), CrdY (Yellow Cards), and CrdR (Red Cards). Then, The data was scaled (standardized) so that all variables contribute equally to the distance calculations.

Clustering:

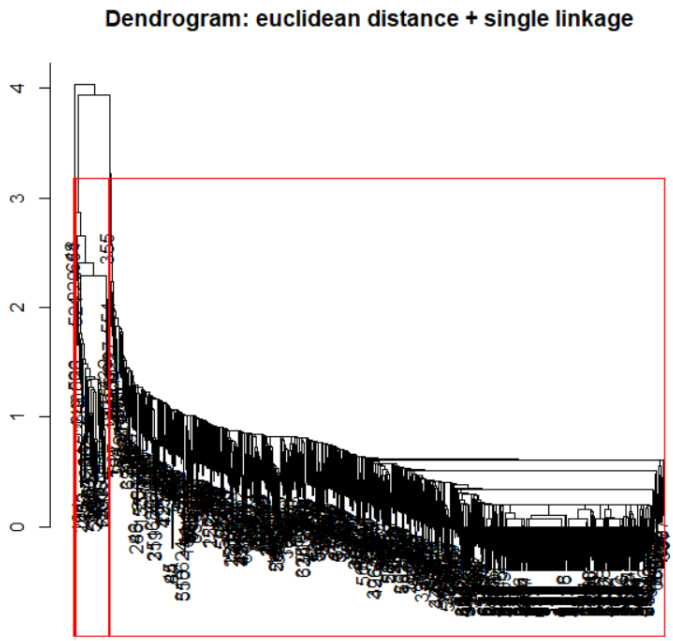
R2 function was written to:

- Compute TSS as the total variance of the entire dataset.
- Compute WSS by calculating the variance within each cluster.
- Return $R^2 = 1 - (WSS/TSS)$, where higher values indicate better cluster fit.

Hierarchical clustering:

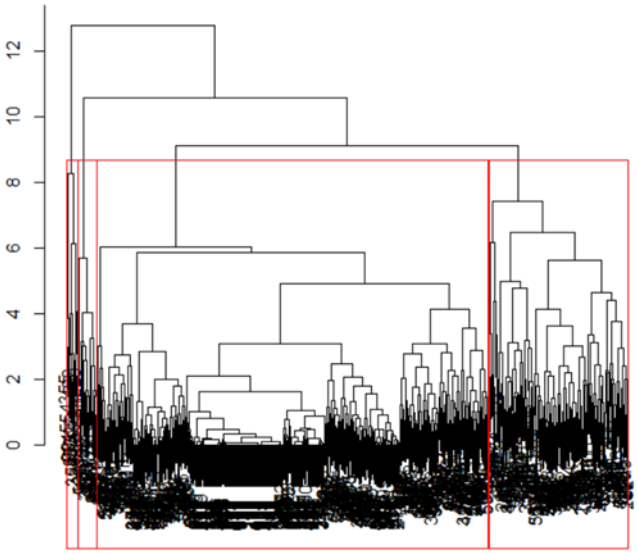
Five different distance metrics were used to compute pairwise distances between observations: Euclidean, Manhattan, Binary, Canberra, and Minkowski. For each distance metric, five linkage methods were applied: Single, Complete, Average, Centroid, and Ward.D2. The number of clusters was fixed at $k = 4$ because there are four different categories for football players: Goalkeepers, Defenders, Midfielders, and Strikers. The size of each cluster and R^2 value was computed for each of the 25 combinations and stored in a results table.

1) Euclidean distance + single linkage:



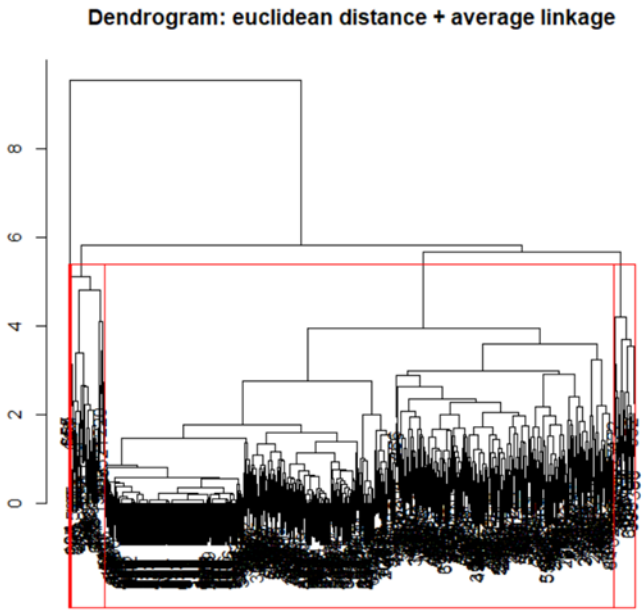

```
=== Analyzing with: euclidean distance and single linkage ===
Cluster sizes:
clusters
  1   2   3   4
645  39   2   1
R-squared: 0.178321
Cluster centers:
          Age          MP      Starts         Min         Gls         Ast        CrdY       CrdR
1 -0.02227222 -0.05451215 -0.05836556 -0.05762833 -0.03542004 -0.04691402 -0.07780366 -0.2462804
2  0.32372414  0.79808822  0.85364455  0.84206942  0.35975368  0.59024392  1.18234386  3.6884783
3  0.41005896  1.25481044  1.43739334  1.37778934  0.82939480  0.48220847  2.22704124  7.6232370
4  0.92021924  1.52527718  1.47886183  1.57398617  7.15674156  6.27560974 -0.38212908 -0.2462804
```

2) Euclidean distance + complete linkage:



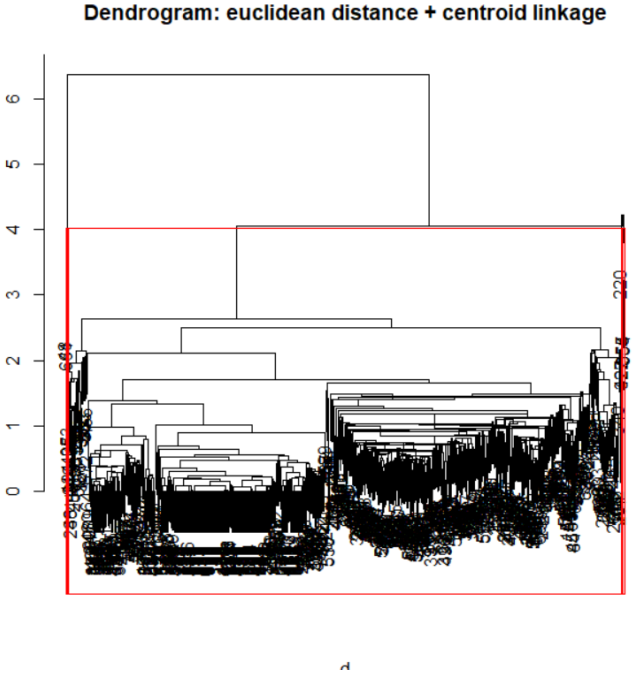

```
=== Analyzing with: euclidean distance and complete linkage ===
Cluster sizes:
clusters
  1   2   3   4
171  23 479  14
R-squared: 0.5018493
Cluster centers:
         Age        MP      Starts         Min         Gls         Ast        CrdY       CrdR
1  0.4404896  1.177308  1.2237457  1.2170280  0.5859604  0.85796336  0.8385354 -0.1542392
2  0.5120910  1.088499  1.2516866  1.2388849  0.1054421  0.09292854  1.6772829  4.0306312
3 -0.2078847 -0.513266 -0.5407144 -0.5377686 -0.3455548 -0.37762068 -0.4064316 -0.1641351
4  0.8910672  1.392804  1.4966340  1.4989301  4.4925956  2.28794393  0.9081200  0.8779364
```

3) Euclidean distance + average linkage



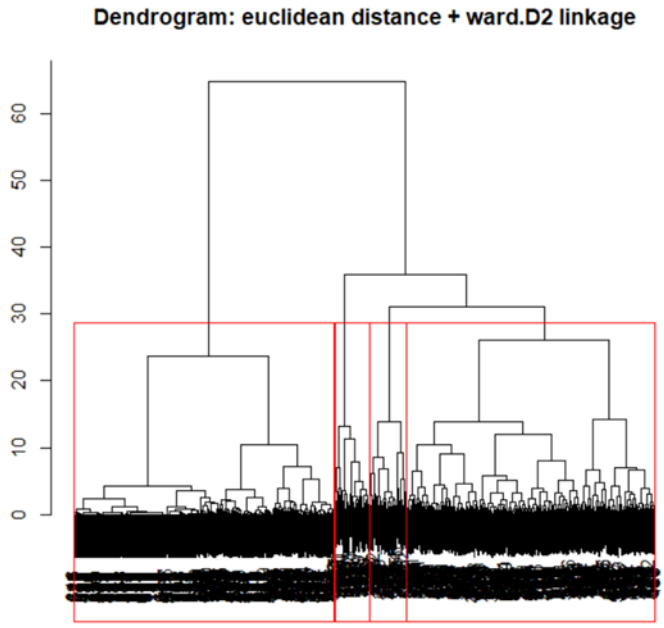

```
=== Analyzing with: euclidean distance and average linkage ===
Cluster sizes:
clusters
  1   2   3   4
 26  41 618   2
R-squared: 0.2854508
Cluster centers:
          Age         MP     Starts        Min        Gls        Ast        CrdY        CrdR
1  0.32372414  1.2458940  1.1439240  1.1367517  2.6609952  2.8117230  0.38981481 -0.2462804
2  0.32793560  0.8203674  0.8821201  0.8682021  0.3826630  0.5849739  1.23330470  3.8804177
3 -0.03835376 -0.1117781 -0.1121058 -0.1108899 -0.1604993 -0.1722976 -0.09763403 -0.2462804
4  0.92021924  1.5252772  1.6862043  1.6890632  7.1567416  4.6955912 -0.18142367 -0.2462804
```

4) Euclidean + centroid linkage



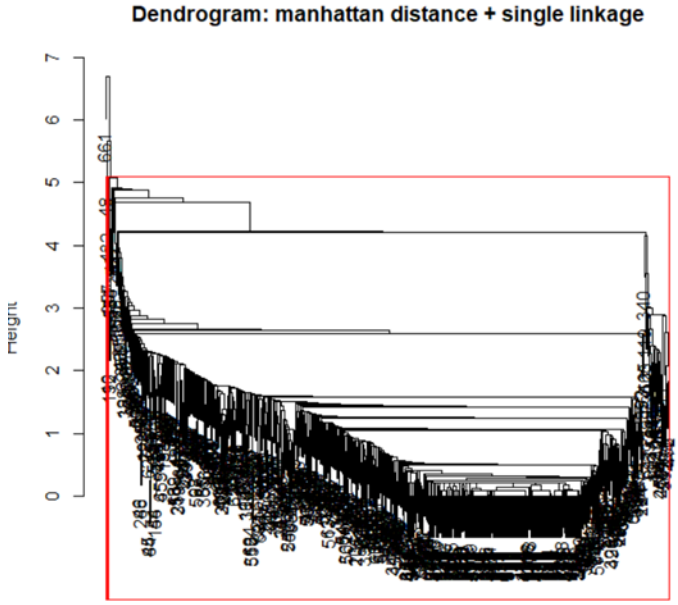

```
=== Analyzing with: euclidean distance and centroid linkage ===
Cluster sizes:
clusters
  1   2   3   4
680   2   1   4
R-squared: 0.07122485
Cluster centers:
           Age          MP      Starts         Min         Gls         Ast         CrdY         CrdR
1 -0.01127341 -0.01297399 -0.01424785 -0.01421087 -0.04428838 -0.01735621 -0.01200469 -0.02639681
2  0.41005896  1.25481044  1.43739334  1.37778934  0.82939480  0.48220847  2.22704124  7.62323698
3  0.71615513  1.37072476  1.56179882  1.57209967  4.15957730 -0.04446437  1.62492501  3.68847830
4  1.53241158  1.23549139  1.31298786  1.33392802  6.07443224  2.72056805  0.52104527 -0.24628038
```

5) Euclidean distance + ward's method:



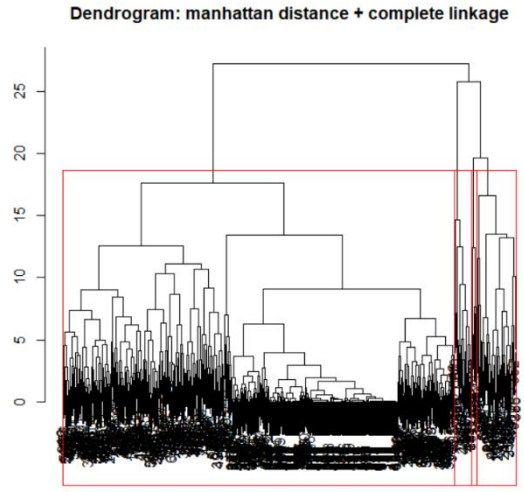

```
=== Analyzing with: euclidean distance and ward.D2 linkage ===
Cluster sizes:
clusters
  1   2   3   4
 44  41 294 308
R-squared: 0.587034
Cluster centers:
          Age         MP     Starts        Min        Gls        Ast       CrdY       CrdR
1  0.1317897  1.3355992  1.3356070  1.3168637  2.75938693  2.0263175  0.9133331 -0.2462804
2  0.3279356  0.8203674  0.8821201  0.8682021  0.38266300  0.5849739  1.2333047  3.8804177
3  0.3357907  0.6910095  0.6128192  0.6261057  0.01497261  0.1848354  0.4357113 -0.2462804
4 -0.3830084 -0.9596046 -0.8931898 -0.9013421 -0.45942908 -0.5437776 -0.7105561 -0.2462804
```

6) Manhattan distance + single linkage:



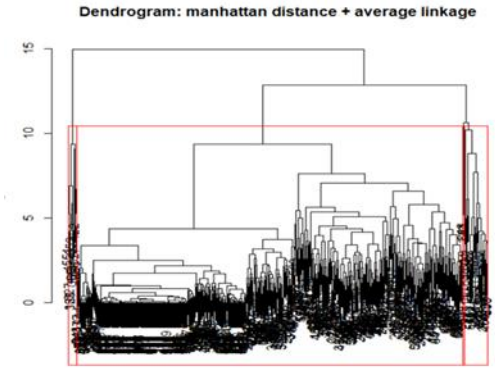

```
=== Analyzing with: manhattan distance and single linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.03296988
Cluster centers:
           Age          MP       Starts          Min          Gls          Ast         CrdY         CrdR
1 -0.002246013 -0.00420432 -0.004667443 -0.004381209 -0.00217709 -0.007504876 -0.009474296 -0.02768268
2 -0.304165430  1.06161991  1.395924844  1.305158775  0.16335830 -0.571137217  1.624925015  7.62323698
3  0.716155129  0.36613401  0.317744028  0.241168226 -0.16965995  4.168918369  2.026335833  3.68847830
4  1.124283352  1.44800097  1.478861830  1.450419897  1.49543130  1.535554155  2.829157469  7.62323698
```

7) Manhattan distance + complete linkage:



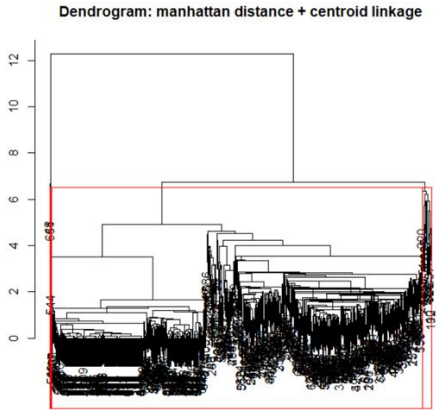

```
=== Analyzing with: manhattan distance and complete linkage ===
Cluster sizes:
clusters
  1   2   3   4
 60  26 593   8
R-squared: 0.3463051
Cluster centers:
          Age         MP     Starts        Min        Gls        Ast       CrdY        CrdR
1  0.12777027  1.3385263  1.3641323  1.3539565  1.4621295  1.8866694  1.1766829  0.01603686
2  0.51209102  1.1805064  1.3512665  1.3362499  0.5860353  0.4214385  1.7175583  3.99115204
3 -0.05123605 -0.2052929 -0.2162415 -0.2145428 -0.2432272 -0.2389692 -0.2020696 -0.17329161
4  1.17529938  1.3417462  1.4062920  1.4054974  5.1586321  2.1938952  0.5712216 -0.24628038
```

8) Manhattan distance + average linkage:



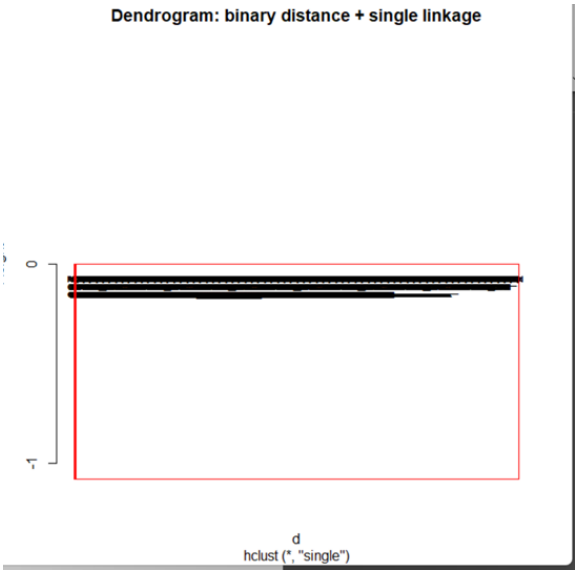

```
=== Analyzing with: manhattan distance and average linkage ===
Cluster sizes:
clusters
  1   2   3   4
 14  39 632   2
R-squared: 0.2499756
Cluster centers:
          Age          MP      Starts         Min        Gls        Ast       CrdY       CrdR
1  0.38090695  1.42040232  1.52033032  1.50169249  3.7551980  3.4541481  0.7934312 -0.2462804
2  0.32372414  0.79808822  0.85364455  0.84206942  0.3597537  0.5902439  1.1823439  3.6884783
3 -0.02971212 -0.08468464 -0.09090435 -0.08958858 -0.1080094 -0.1144652 -0.0975847 -0.2462804
4  0.41005896  1.25481044  1.43739334  1.37778934  0.8293948  0.4822085  2.2270412  7.6232370
```

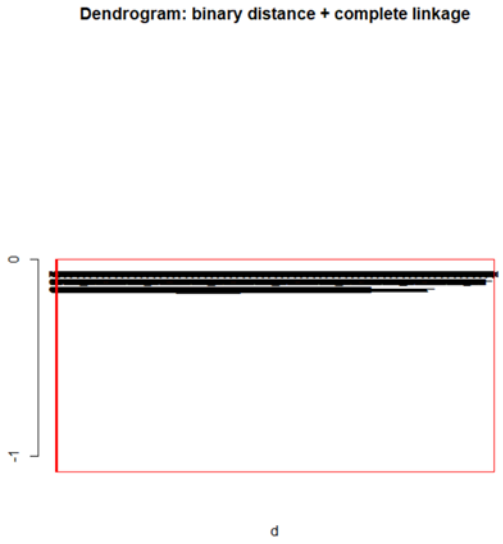
9) Manhattan distance + centroid linkage:



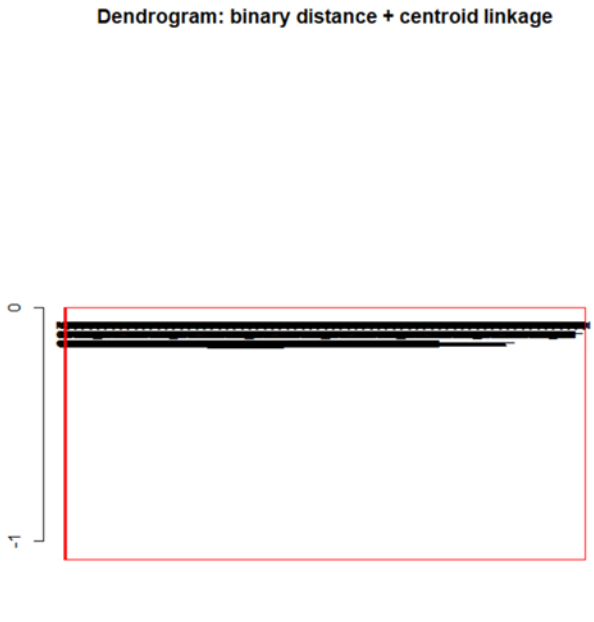

```
=== Analyzing with: manhattan distance and centroid linkage ===
Cluster sizes:
clusters
  1   2   3   4
668  17   1   1
R-squared: 0.1179145
Cluster centers:
          Age         MP      Starts         Min         Gls         Ast        CrdY        CrdR
1 -0.01334352 -0.0334349 -0.03597973 -0.03551303 -0.08092156 -0.08625129 -0.02999324 -0.06367931
2  0.47607970  1.1661701  1.24468681  1.23336050  3.08216532  3.33243797  0.91655298  1.60537076
3 -0.30416543  1.0616199  1.39592484  1.30515878  0.16335830 -0.57113722  1.62492501  7.62323698
4  1.12428335  1.4480010  1.47886183  1.45041990  1.49543130  1.53555415  2.82915747  7.62323698
```

10) Binary distance:










```
=== Analyzing with: binary distance and single linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.003071306
Cluster centers:
          Age          MP       Starts          Min          Gls          Ast         CrdY
1  0.004019113  0.005172764  0.004426525  0.004523164  0.002204729  0.002504988  0.003436578
2 -0.916357765 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
3 -1.120421877 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
4 -0.712293653 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
         CrdR
1  0.001080177
2 -0.246280383
3 -0.246280383
4 -0.246280383
```

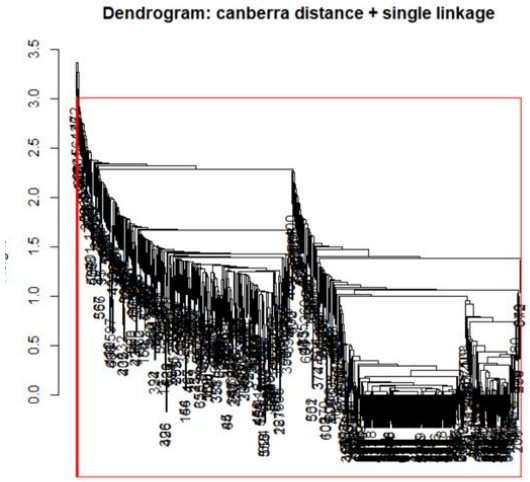
```
=== Analyzing with: binary distance and complete linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.003071306
Cluster centers:
          Age          MP       Starts          Min          Gls          Ast         CrdY
1  0.004019113  0.005172764  0.004426525  0.004523164  0.002204729  0.002504988  0.003436578
2 -0.916357765 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
3 -1.120421877 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
4 -0.712293653 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
         CrdR
1  0.001080177
2 -0.246280383
3 -0.246280383
4 -0.246280383
```

```
=== Analyzing with: binary distance and average linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.003071306
Cluster centers:
          Age          MP       Starts          Min          Gls          Ast         CrdY
1  0.004019113  0.005172764  0.004426525  0.004523164  0.002204729  0.002504988  0.003436578
2 -0.916357765 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
3 -1.120421877 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
4 -0.712293653 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
         CrdR
1  0.001080177
2 -0.246280383
3 -0.246280383
4 -0.246280383
```

```
=== Analyzing with: binary distance and centroid linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.003071306
Cluster centers:
          Age          MP       Starts          Min          Gls          Ast         CrdY
1  0.004019113  0.005172764  0.004426525  0.004523164  0.002204729  0.002504988  0.003436578
2 -0.916357765 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
3 -1.120421877 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
4 -0.712293653 -1.179390210 -1.009247747 -1.031281466 -0.502678203 -0.571137217 -0.783539894
         CrdR
1  0.001080177
2 -0.246280383
3 -0.246280383
4 -0.246280383
```

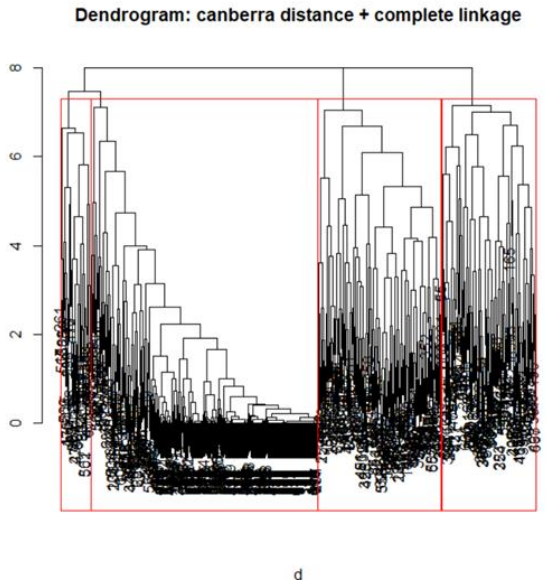
In the dendrograms above, there is a clear problem with the results of binary hierarchical clustering. All the clusters are joined at almost the same level, and the tree is very flat. This means the clustering is not working properly. The error is likely because binary distance was used with data that is not binary (not just 0s and 1s). Binary distance is only useful when we have data that shows presence or absence (like yes/no or true/false). If we use it with normal numerical data, the results become meaningless, and the clustering fails.

11) Canberra + single linkage:



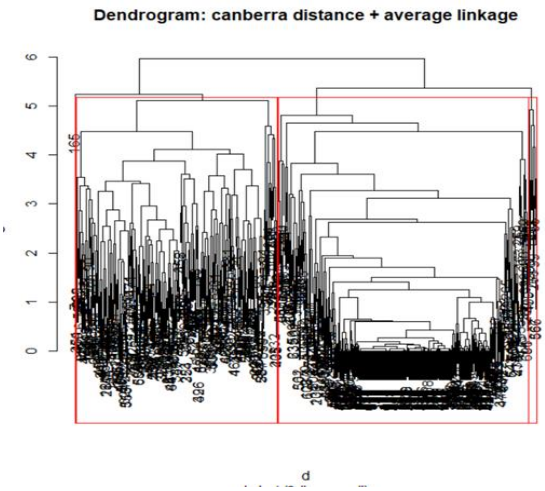

```
=== Analyzing with: canberra distance and single linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.003523115
Cluster centers:
           Age            MP        Starts           Min          Gls          Ast          CrdY
1 -0.0004559772 -0.0000241741 -5.983222e-05  0.0003074807  0.001230991 -0.002884939 -8.456905e-05
2 -0.3041654299  0.2115815912 -1.400392e-02 -0.0918004482  0.163358298  1.535554155  1.928174e-02
3  1.3283474636 -0.1747994645 -1.400392e-02 -0.0748218756 -0.502678203 -0.044464374  1.928174e-02
4 -0.7122936533 -0.0202470422  6.893307e-02 -0.0436944925 -0.502678203  0.482208469  1.928174e-02
          CrdR
1 -0.004672394
2 -0.246280383
3  3.688478299
4 -0.246280383
```

12) Canberra distance + complete linkage:



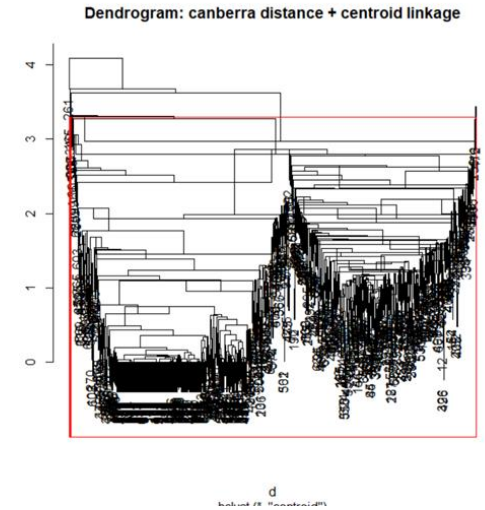

```
=== Analyzing with: canberra distance and complete linkage ===
Cluster sizes:
clusters
  1   2   3   4
179 137  43 328
R-squared: 0.4831926
Cluster centers:
         Age         MP     Starts        Min        Gls        Ast        CrdY       CrdR
1  0.3775683  1.2416433  1.3787814  1.3724723  0.6991642  0.6969968  0.99029227  0.4791221
2  0.2886632  0.5731440  0.3661744  0.3835858  0.2460052  0.3591900  0.28884229 -0.2175595
3  0.4456515 -0.1406542 -0.1702341 -0.1732493 -0.3168076 -0.2281875 -0.08340475  0.1197437
4 -0.3850445 -0.8985571 -0.8830723 -0.8865064 -0.4427755 -0.5004860 -0.65014423 -0.1862993
```

13) Canberra distance + average linkage:



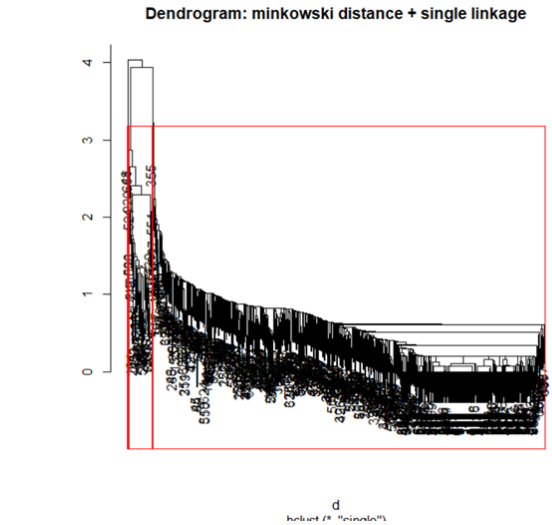

```
=== Analyzing with: canberra distance and average linkage ===
Cluster sizes:
clusters
  1   2   3   4
300 373  13   1
R-squared: 0.4352878
Cluster centers:
         Age         MP     Starts        Min        Gls        Ast        CrdY       CrdR
1  0.3590429  0.9943896  0.9925746  0.9942591  0.5474393  0.58578746  0.72710282  0.1865431
2 -0.3140130 -0.8095831 -0.7951236 -0.7975593 -0.4267893 -0.46664984 -0.59413425 -0.2040846
3  0.7946413  0.2591362 -0.1288397 -0.1024665 -0.4002110 -0.08497767  0.26630378  1.2670883
4 -0.9163578  0.2888578  0.4836180  0.5439528  0.1633583 -0.57113722  0.01928174  3.6884783
```

14) Canberra distance + centroid linkage:



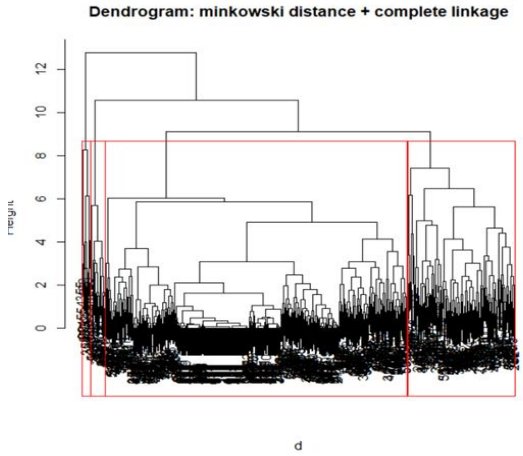

```
=== Analyzing with: canberra distance and centroid linkage ===
Cluster sizes:
clusters
  1   2   3   4
684   1   1   1
R-squared: 0.008395332
Cluster centers:
           Age            MP        Starts           Min           Gls           Ast          CrdY
1  0.0004390409 -0.0002501279  0.0001826736  0.0002026747  0.0002572538 -0.0005749704  0.0005022889
2 -0.9163577650  0.2888578024  0.4836179994  0.5439527704  0.1633582975 -0.5711372168  0.0192817424
3 -0.1001013183 -0.0202470422 -0.0969409020 -0.0710488595 -0.5026782034 -0.0444643739  0.4206925605
4  0.7161551285 -0.0975232534 -0.5116258316 -0.6115334207  0.1633582975  1.0088813118 -0.7835398939
        CrdR
1 -0.01617754
2  3.68847830
3  3.68847830
4  3.68847830
```

15) Minkowski distance + single linkage:



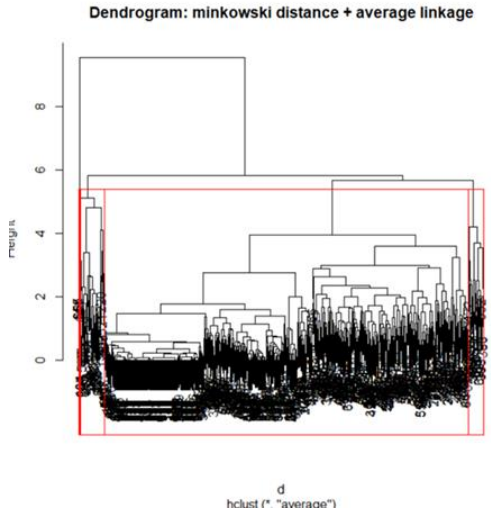

```
=== Analyzing with: minkowski distance and single linkage ===
Cluster sizes:
clusters
  1   2   3   4
645  39   2   1
R-squared: 0.178321
Cluster centers:
          Age          MP      Starts         Min         Gls         Ast        CrdY       CrdR
1 -0.02227222 -0.05451215 -0.05836556 -0.05762833 -0.03542004 -0.04691402 -0.07780366 -0.2462804
2  0.32372414  0.79808822  0.85364455  0.84206942  0.35975368  0.59024392  1.18234386  3.6884783
3  0.41005896  1.25481044  1.43739334  1.37778934  0.82939480  0.48220847  2.22704124  7.6232370
4  0.92021924  1.52527718  1.47886183  1.57398617  7.15674156  6.27560974 -0.38212908 -0.2462804
```

16) Minkowski distnce + complete linkage:



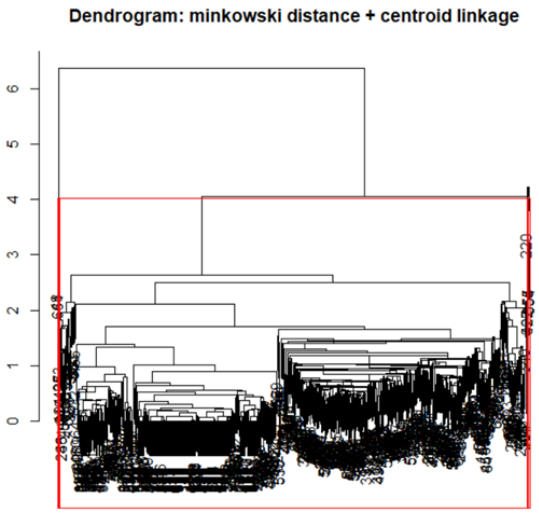

```
=== Analyzing with: minkowski distance and complete linkage ===
Cluster sizes:
clusters
  1   2   3   4
171  23 479  14
R-squared: 0.5018493
Cluster centers:
          Age         MP     Starts        Min        Gls        Ast        CrdY       CrdR
1  0.4404896  1.177308  1.2237457  1.2170280  0.5859604  0.85796336  0.8385354 -0.1542392
2  0.5120910  1.088499  1.2516866  1.2388849  0.1054421  0.09292854  1.6772829  4.0306312
3 -0.2078847 -0.513266 -0.5407144 -0.5377686 -0.3455548 -0.37762068 -0.4064316 -0.1641351
4  0.8910672  1.392804  1.4966340  1.4989301  4.4925956  2.28794393  0.9081200  0.8779364
```

17) Minkowski distance + average linkage:




```
=== Analyzing with: minkowski distance and average linkage ===
Cluster sizes:
clusters
  1   2   3   4
 26  41 618   2
R-squared: 0.2854508
Cluster centers:
           Age          MP      Starts         Min         Gls         Ast         CrdY        CrdR
1  0.32372414  1.2458940  1.1439240  1.1367517  2.6609952  2.8117230  0.38981481 -0.2462804
2  0.32793560  0.8203674  0.8821201  0.8682021  0.3826630  0.5849739  1.23330470  3.8804177
3 -0.03835376 -0.1117781 -0.1121058 -0.1108899 -0.1604993 -0.1722976 -0.09763403 -0.2462804
4  0.92021924  1.5252772  1.6862043  1.6890632  7.1567416  4.6955912 -0.18142367 -0.2462804
```

18) Minkowski distance + centroid linkage:

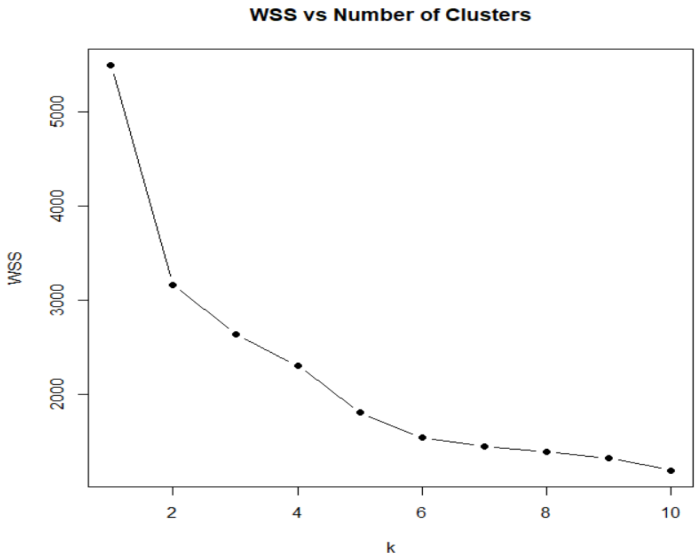



```
=== Analyzing with: minkowski distance and centroid linkage ===
Cluster sizes:
clusters
  1   2   3   4
680   2   1   4
R-squared: 0.07122485
Cluster centers:
           Age          MP      Starts         Min         Gls         Ast         CrdY        CrdR
1 -0.01127341 -0.01297399 -0.01424785 -0.01421087 -0.04428838 -0.01735621 -0.01200469 -0.02639681
2  0.41005896  1.25481044  1.43739334  1.37778934  0.82939480  0.48220847  2.22704124  7.62323698
3  0.71615513  1.37072476  1.56179882  1.57209967  4.15957730 -0.04446437  1.62492501  3.68847830
4  1.53241158  1.23549139  1.31298786  1.33392802  6.07443224  2.72056805  0.52104527 -0.24628038
```

The results of the hierarchical clustering showed that The best result was achieved with Euclidean distance + Ward method linkage ($R^2 = 0.587$). The clustering results of the Ward's method into 4 clusters that had 44, 41, 294, 308 observations respectively. On the other hand, Binary distance consistently produced extremely low R^2 values (≈0.002), indicating that it is unsuitable for this dataset. Additionally, centroid linkage tended to perform poorly across most distance methods.

K-means clustering:

Since k-means requires Euclidean distance, it was applied to the scaled data using only that metric. The number of clusters (k) was varied from 2 to 10. An elbow graph was created between the number of clusters and within-cluster sum of squares (WSS). The elbow point is at k = 4 because this is the point beyond which adding more clusters doesn't result in much better modeling of the data.



The results showed that K-means provided higher R^2 values than hierarchical clustering for the same k. Additionally, R^2 for K-means increased consistently with more clusters. This shows that adding more clusters allows k-means to explain more variance in the data, though it may also risk overfitting if k becomes too large.

We compared the performance of hierarchical clustering (using Ward's method with Euclidean distance) and K-means clustering with 4 clusters. The results show that K-means performed better, with an R-squared value of 0.697, compared to 0.587 for hierarchical clustering. This means that K-means explained more of the variation in the data, resulting in a better overall clustering fit.

K-means gave a slightly more balanced distribution than hierarchical clustering, especially by creating a third medium-sized cluster instead of just two very large ones. Overall, K-means not only provided a better fit but also more practical cluster sizes for further analysis.